

Agentic Coding Overview

Models, harnesses, access, routing, and benchmarks

Cédric Jung Jozef Janus

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Model	Context	Deployment note
DeepSeek V4 Flash 284B	393k tokens	RTX PRO 6000 Blackwell migration
Qwen 3.6 35B	262k tokens	4 replicas, one A40 each
GLM 5.2 744B Preview	262k tokens	8 AMD MI300X GPUs

Transition

Qwen 3.5 397B reached its announced retirement date on 30 September. DeepSeek V4 Flash is its planned replacement on the Blackwell infrastructure.

AQUEDUCT: what may come next

Release-note candidates are not confirmed deployments

Candidate model	Discussed hardware	Status
Qwen3.8-27B	freed H200 node; possibly 4 instances	Candidate
DeepSeek V4 Flash Vision Exp	RTX PRO 6000 Blackwell	Candidate
GLM-5.3-Flash	RTX PRO 6000 Blackwell	Candidate
DeepSeek V4.1 Flash / GLM-5.3	AMD MI300X	Candidate

Before use

Check the live `/v1/models` endpoint. A release-note mention does not make a model available through AQUEDUCT.

Hardware behind the AQUEDUCT models

Memory and replica counts shape what can be served

Accelerator	Memory / GPU	AQUEDUCT use
NVIDIA A40	48 GB	Qwen 3.6 35B: 4 independent one-GPU replicas
RTX PRO 6000 Blackwell	96 GB	DeepSeek V4 Flash replaces the retiring Qwen 3.5 deployment
NVIDIA H200	141 GB	Initially served DeepSeek; freed for a future deployment
AMD MI300X	192 GB	GLM 5.2 744B Preview: 8 GPUs in one replica

Hardware mapping reflects the public overview plus migration notes; verify the live deployment before relying on it.

Sources: TU Wien: [AQUEDUCT overview](#); [release notes](#)

1. **Request access** through the TU Wien AQUEDUCT access process.
2. **Wait for approval / allowlisting** before using the service.
3. **Use the issued credentials** in an API-compatible client or coding harness.
4. **Check available models** through the live `/v1/models` endpoint.

Practical point

The public model catalog describes the service; it does not grant an account automatically.

Model: decides

- Understands the task and code
- Proposes edits and tool calls
- Interprets tool results

Examples: GPT, Claude, Qwen, DeepSeek

Harness: executes the loop

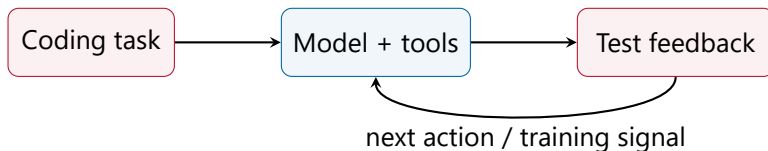
- Builds context and exposes tools
- Reads files, edits code, runs tests
- Manages state and permissions

Examples: Codex, Claude Code, OpenCode

User task → harness ↔ model → tested change

Source: [OpenAI](#), *Unrolling the Codex agent loop*

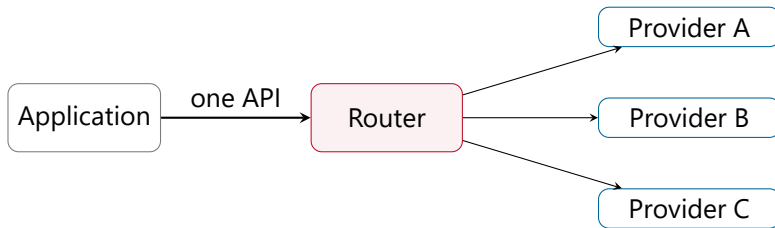
Codex as a concrete example



- Coding post-training can use trajectories: inspect, edit, run tests, observe, revise.
- OpenAI describes codex-1 and GPT-5-Codex as trained or optimized for agentic coding in tool-using environments.
- A benchmark therefore measures the **model, harness, tools, and reasoning budget** together.

Why LLM routing platforms exist

One integration, several model and provider choices



- **Model** = which neural network responds.
- **Provider** = who runs it; providers vary in price, speed, region, and retention rules.
- The router can select a provider, apply constraints, and fail over when needed

What it offers

- One API across many models
- Provider ordering and filtering
- Price or throughput priority
- Provider and model fallback

Routing choices

- Cheapest eligible endpoint
- Higher throughput endpoint
- Specific provider or region
- EU in-region option for eligible enterprise use

Check the chosen provider and region before sending sensitive code or data.

Sources: [OpenRouter quickstart](#); [sovereign AI docs](#)

Different ways to constrain where inference runs

Service	Role	Regional control
EUrouter	Multi-provider model gateway	EU processing by design; provider filters
Eden AI	Multi-provider AI gateway	Dedicated European endpoint
Scaleway	Inference provider with many models	European-hosted serving; not a multi-provider router

Selection question

Specify processing location, provider ownership, and retention requirements separately. "EU-hosted" and "EU-owned" are different constraints.

Subscription: fixed monthly fee

ChatGPT Plus: \$20/month

Codex usage included, subject to plan limits.

API: metered tokens

GPT-5.6 Sol: \$4/M input + \$20/M output

5M input + 1M output = \$40.

For frontier models, a sufficiently active developer can therefore consume **more than \$20 of API tokens very quickly**, while the same \$20 subscription may continue to provide included usage until its rate or weekly limits are reached.

At the moment, subscription pricing can offer dramatically better effective value for a human using an AI coding agent interactively than paying standard API rates for every token.

Sources: [ChatGPT Plus](#); [Codex plan usage](#); [API pricing](#)

Which benchmarks answer which questions?

One result slide per benchmark follows

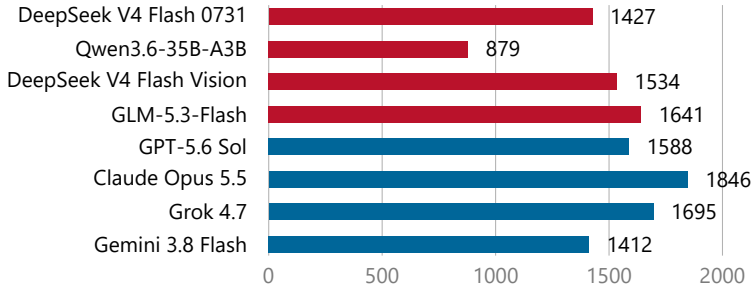
Benchmark	Metric	What it measures
GDPval-AA v2.1	Elo	Professional written knowledge work (writing proxy)
DeepSWE v1.1	Resolved %	Long-horizon software engineering
Terminal-Bench 4.0	Pass rate %	Agentic terminal tasks; common evaluator
Terminal-Bench 2.1	Pass rate %	Older version with more open-model coverage
Humanity's Last Exam	pass@1 %	Broad expert academic reasoning
GPQA Diamond	Accuracy %	Graduate-level physics, biology, chemistry

Charts: red =

open-weight comparison models; blue = proprietary models. Compare exact versions; N/A means no public exact-model result, not zero.

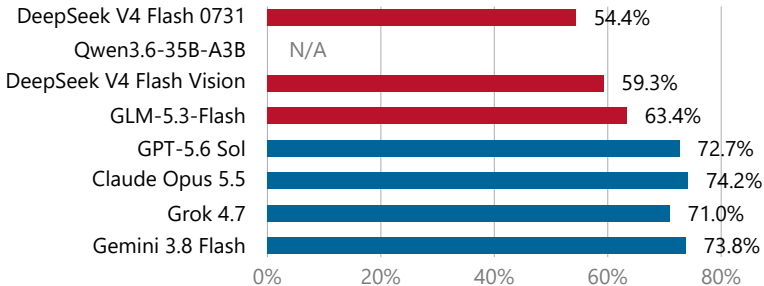
Sources: [Terminal-Bench](#); [DeepSWE](#); [Artificial Analysis](#)

Professional written knowledge-work proxy; Elo, higher is better



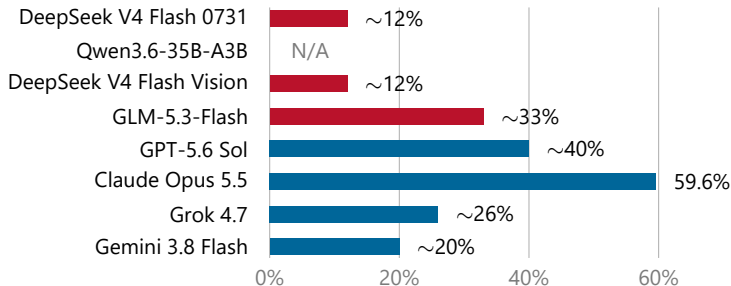
Same benchmark version and evaluator across all eight models. This is not a direct academic-paper-writing test.

Source: [Artificial Analysis](#), GDPval-AA v2.1



Qwen: no public exact-model v1.1 result. Harnesses and reasoning budgets differ across reported scores.

Sources: [DeepSWE leaderboard](#); [DeepSeek Vision card](#); [GLM Flash card](#); [Gemini card](#)

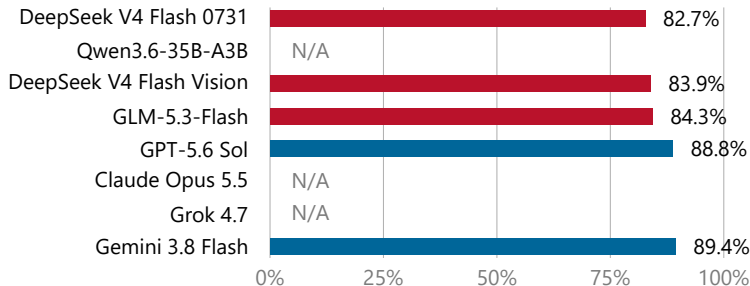


Qwen: no exact v4.0 score. Scores remain sensitive to harness, tools, and reasoning settings.

Source: [Artificial Analysis, Terminal-Bench 4.0](#)

Terminal-Bench 2.1

Agentic terminal tasks; published results under different setups

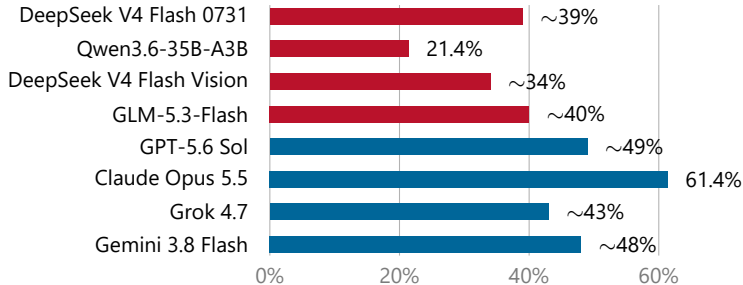


Several exact-model scores are unavailable. Do not compare v2.1 percentages with v4.0 percentages.

Sources: [Terminal-Bench](#); [DeepSeek Vision card](#); [GLM Flash card](#); [Gemini card](#)

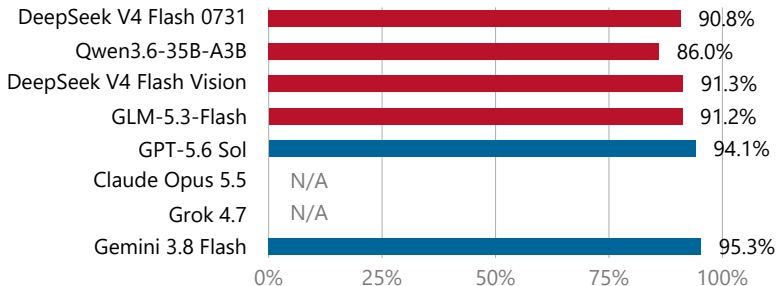
Humanity's Last Exam

Expert academic reasoning; text-only subset pass@1, higher is better



Artificial Analysis uses a 2,158-question text-only subset. HLE spans humanities and mathematics as well as science.

Sources: [Artificial Analysis HLE](#); [Qwen model card](#)



198 questions in physics, biology, and chemistry. GPT-5.6 Sol value is the Artificial Analysis result.

Sources: [Artificial Analysis GPQA](#); [Qwen model card](#)

- [TU Wien AQUEDUCT overview and release notes](#)
- [OpenAI: Unrolling the Codex agent loop](#)
- [OpenAI: GPT-5-Codex system-card addendum](#)
- [OpenRouter quickstart and regional routing](#)
- [EUrouter documentation, Eden AI EU, Scaleway FAQ](#)
- [ChatGPT Plus and GPT-5.6 Sol API pricing](#)

Deck compiled from the repository's [documentation](#)/ [research notes](#), dated 30 September 2026. Links are clickable in the PDF.

- Artificial Analysis: GDPval-AA v2.1
- DataCurve: DeepSWE v1.1
- Artificial Analysis: Terminal-Bench 4.0
- Artificial Analysis: Humanity's Last Exam
- Artificial Analysis: GPQA Diamond
- Terminal-Bench project

Interpretation

Agentic scores reflect model, harness, tools, and reasoning budget. Benchmark versions and evaluators must match for a fair comparison.